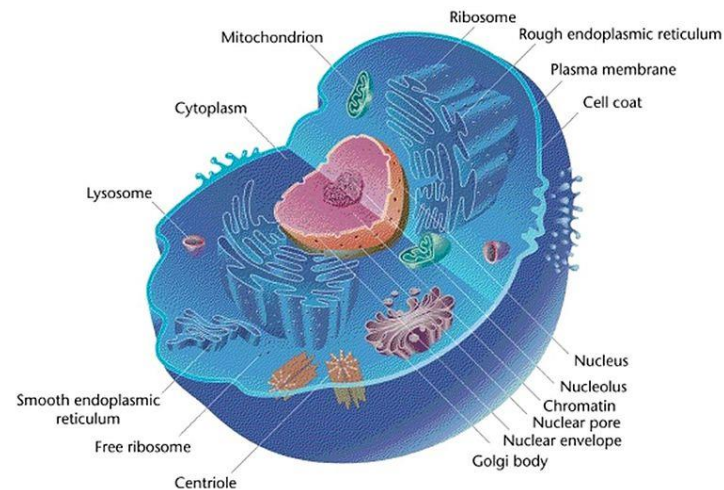


Lysosomal Storage Diseases

Jason Ryan, MD, MPH

Lysosomes

- Membrane-bound organelles of cells
- Contain enzymes
- Breakdown numerous biological structures
 - Proteins, nucleic acids, carbohydrates, lipids
- Digest obsolete components of the cell

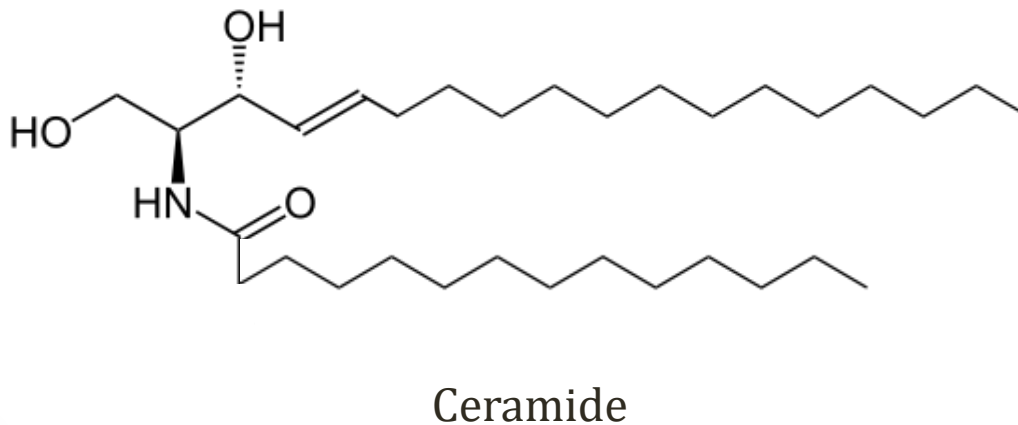
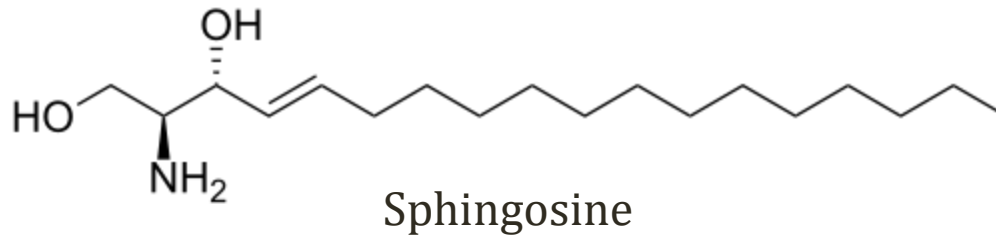


Lysosomal Storage Diseases

- Absence of lysosomal enzyme
- Inability to breakdown complex molecules
- Accumulation → disease
- Most autosomal recessive
- Most have no treatment or cure

Sphingolipids

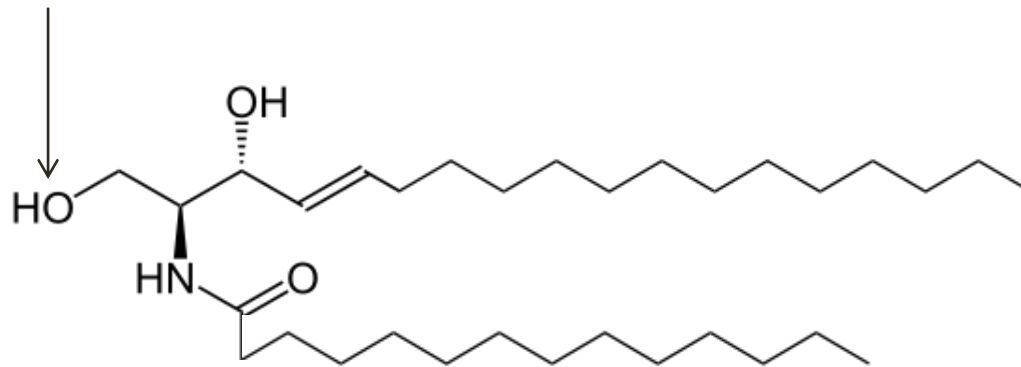
- **Sphingosine**: long chain “amino alcohol”
- Addition of **fatty acid** to NH_2 = **Ceramide**



Ceramide Derivatives

- Modification of “head group” on ceramide
- Yields glycosphingolipids, sulfatides, others
- Very important structures for **nerve tissue**
- Lack of breakdown → accumulation **liver**, **spleen**

Head Group

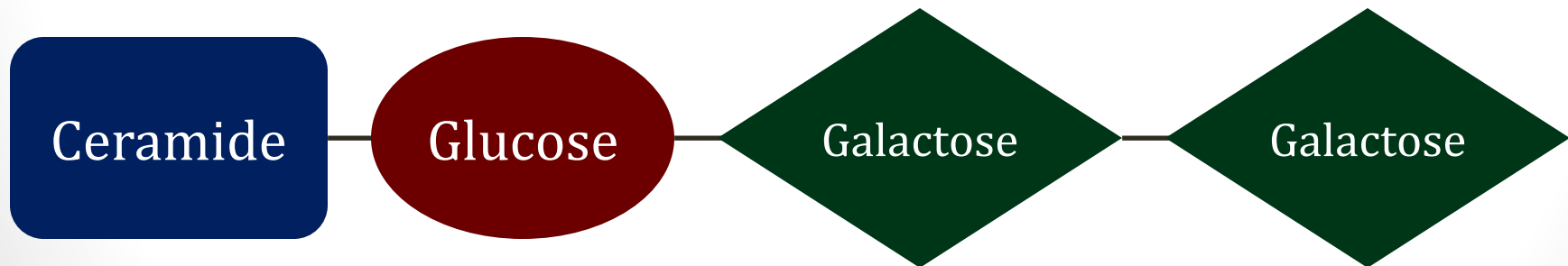


Ceramide

Ceramide Trihexoside

Globotriaosylceramide (Gb3)

- Three sugar head group on ceramide
- Broken down by **α -galactosidase A**
- **Fabry's Disease**
 - Deficiency of α -galactosidase A
 - Accumulation of ceramide trihexoside



Fabry's Disease

- **X-linked recessive** disease
- Slowly progressive symptoms
- Begins child → early adulthood

Fabry's Disease

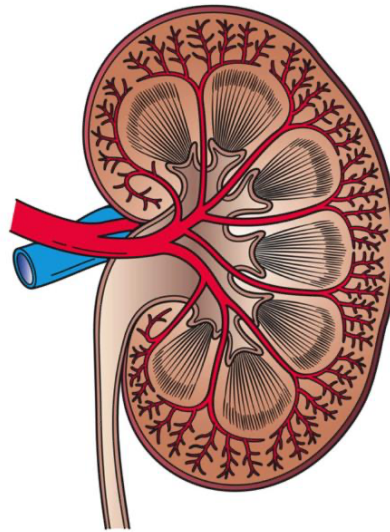
- Neuropathy
 - Classically pain in limbs, **hands, feet**
- Skin: **angiokeratomas**
 - Small dark, red to purple raised spots
 - Dilated surface capillaries
- **Decreased sweat**



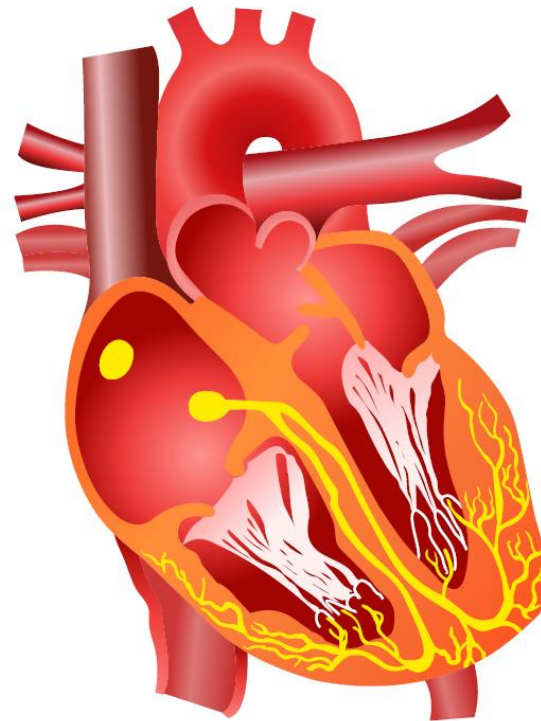
Ldmochowski/Wikipedia

Fabry's Disease

- Renal disease
 - Proteinuria, renal failure
- Cardiac disease
 - Left ventricular hypertrophy
 - Heart failure

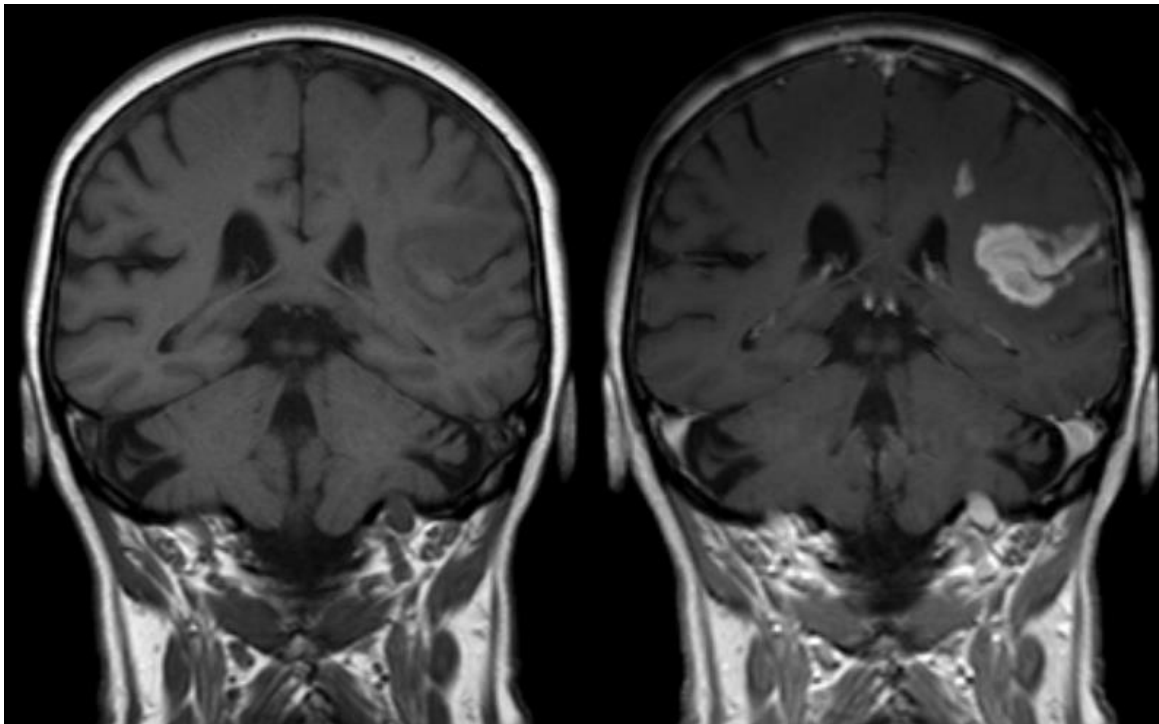


Holly Fischer/Wikipedia



Fabry's Disease

- CNS problems
 - TIA/Stroke (early age)



[Hellerhoff/Wikipedia](#)

Fabry's Disease

- Often misdiagnosed initially
- Enzyme replacement therapy available
 - Recombinant galactosidase

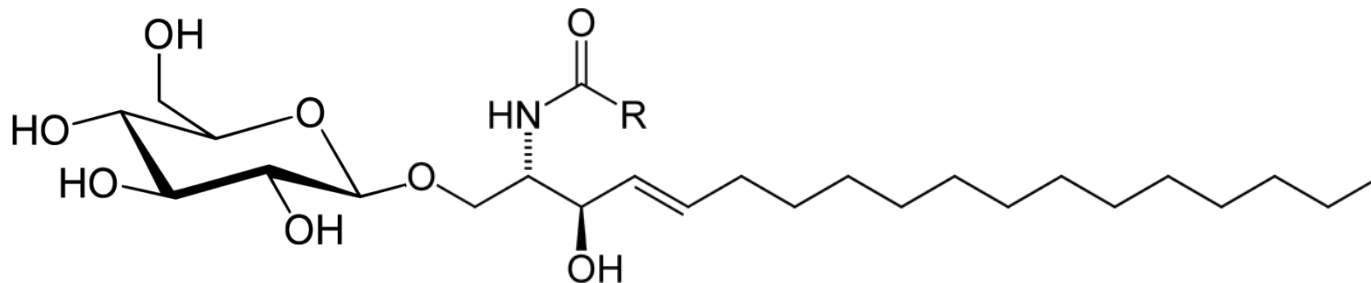
Fabry's Disease

- Classic case
 - Child with pain in hands/feet
 - Lack of sweat
 - Skin findings

Deficiency of α -galactosidase A
Accumulation of ceramide trihexoside

Glucocerebroside

- Glucose head group on ceramide
- Broken down by **glucocerebrosidase**
- **Gaucher's disease**
 - Deficiency of glucocerebrosidase
 - Accumulation of glucocerebroside



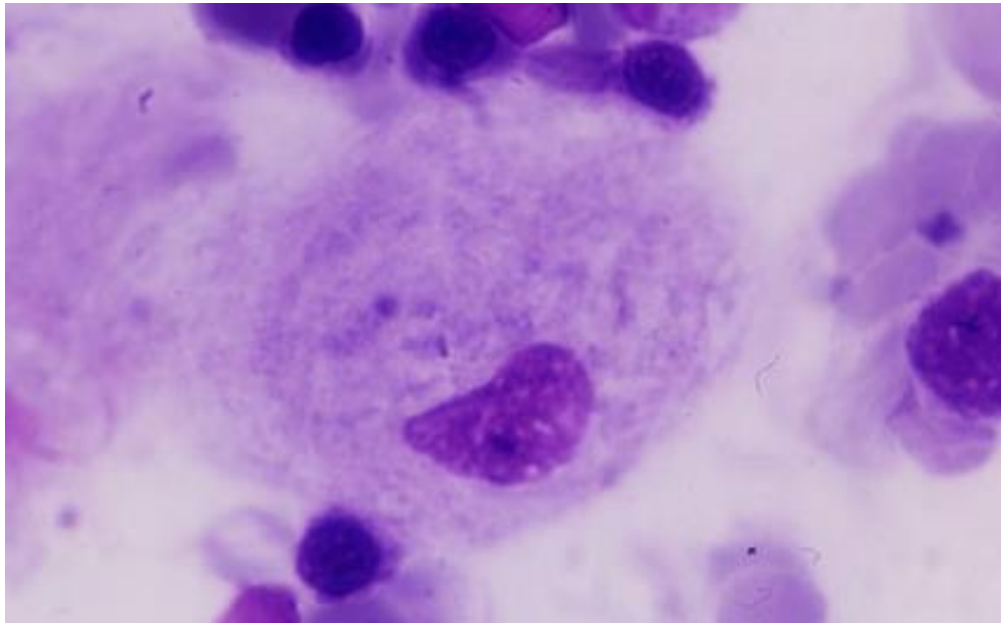
Gaucher's Disease

- Most common lysosomal storage disease
- Autosomal recessive
- More common among **Ashkenazi Jewish** population
- Lipids accumulate in spleen, liver, bones

Gaucher's Disease

- Hepatosplenomegaly:
 - **Splenomegaly**: most common initial sign
- Bones
 - Marrow: Anemia, thrombocytopenia, rarely leukopenia
 - Often **easy bruising** from low platelets
 - Avascular necrosis of joints (joint collapse)
- CNS (rare, neuropathic forms of disease)
 - Gaze palsy
 - Dementia
 - Ataxia

Gaucher's Disease



Gaucher Cell: **Macrophage** filled with lipid
“Crinkled paper”

Bone Crises

- Severe bone pain
- Due to bone infarction (ischemia)
- Infiltration of Gaucher cells in intramedullary space
- Intense pain, often with fever (like sickle cell)



Scuba-limp/Wikipedia

Gaucher's Disease

- Type I
 - Most common form
 - Presents childhood to adult
 - Minimal CNS dysfunction
 - Hepatosplenomegaly, bruising, anemia, joint problems
 - Normal lifespan possible
 - Enzyme replacement therapy
 - Synthetic (recombinant DNA) glucocerebrosidase

Gaucher's Disease

- Type II
 - Presents in infancy with marked CNS symptoms
 - Death <2yrs
- Type III
 - Childhood onset; progressive dementia; shortened lifespan

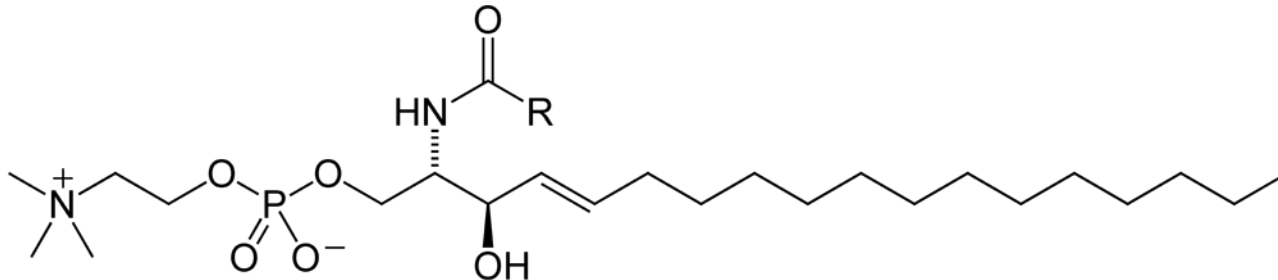
Gaucher's Disease

- Classic case:
 - Child of Ashkenazi Jewish descent
 - Splenomegaly on exam
 - Anemia
 - Bruising (low platelets)
 - Joint pain/fractures

Deficiency of glucocerebrosidase
Accumulation of glucocerebroside

Sphingomyelin

- Phosphate-nitrogen head group
- Broken down by **sphingomyelinase**
- **Niemann-Pick disease**
- Deficiency of acid sphingomyelinase (ASM)
 - Accumulation of sphingomyelin

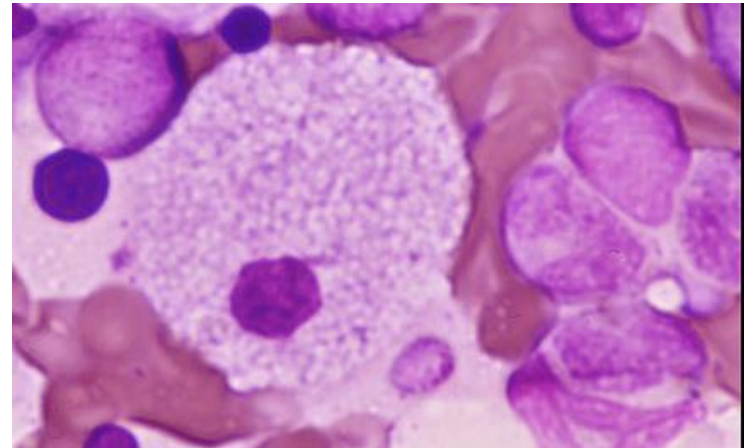


Niemann-Pick Disease

- Autosomal recessive
- More common among **Ashkenazi Jewish** population
- Splenomegaly, neurologic deficits
- Multiple subtypes of disease
- Presents in infancy to adulthood based on type

Niemann-Pick Disease

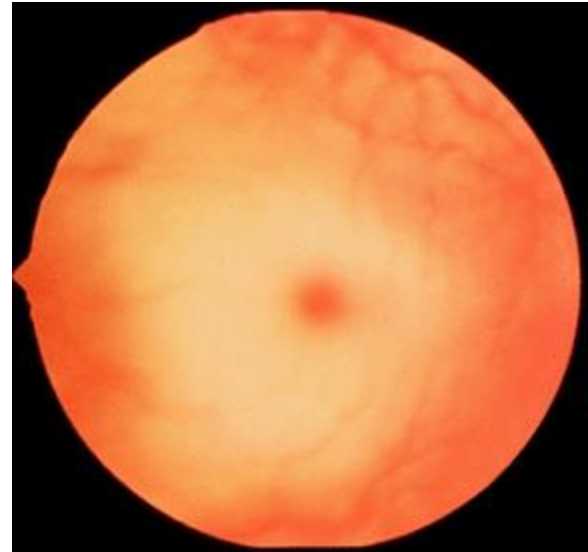
- **Hepatosplenomegaly**
 - 2° thrombocytopenia
- Progressive neuro impairment
 - Weakness: will worsen over time
 - Classic presentation: child that **loses motor skills**
- Pathology
 - Large macrophages with lipids
 - “**Foam cells**”
 - Spleen, bone marrow
- Severe forms: death <3-4yrs



www.hematologistatlas.com; used with permission

Cherry Red Spot

- Seen in many conditions:
 - Niemann-Pick
 - Tay-Sachs
 - Central retinal artery occlusion



Jonathan Trobe, M.D./Wikipedia

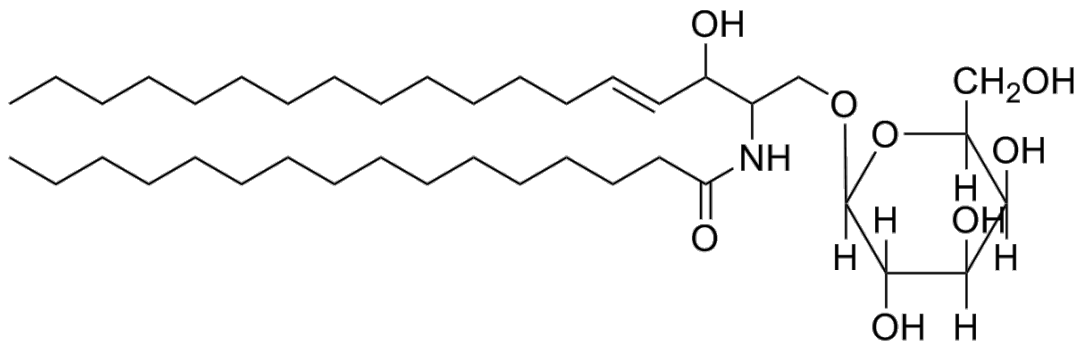
Niemann-Pick Disease

- Classic case:
 - Previously well, healthy child
 - Weakness, loss of motor skills
 - Enlarged liver or spleen on physical exam
 - Cherry red spot

Deficiency of sphingomyelinase
Accumulation of sphingomyelin

Galactocerebroside

- Galactose head group
- Broken down by **galactocerebrosidase**
- Major component of **myelin**
- **Krabbe's Disease**
 - Deficiency of galactocerebrosidase
 - Abnormal metabolism of galactocerebroside

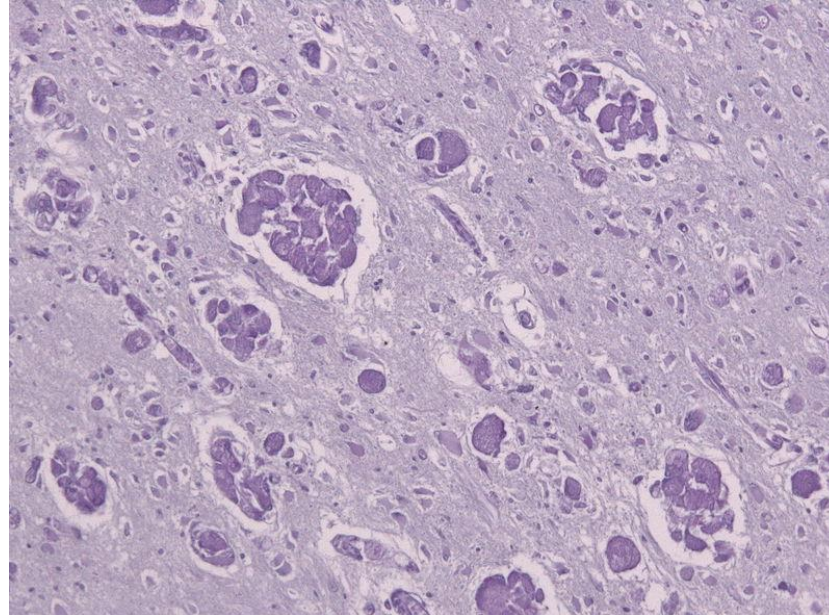


Krabbe's Disease

- Autosomal Recessive
- Usually presents <6 months of age
- Only neuro symptoms
- **Progressive weakness**
 - Developmental delay
 - Eventually floppy limbs, loss of head control
- Absent reflexes
- Optic atrophy: vision loss
- Often fever without infection
- Usually death <2 years

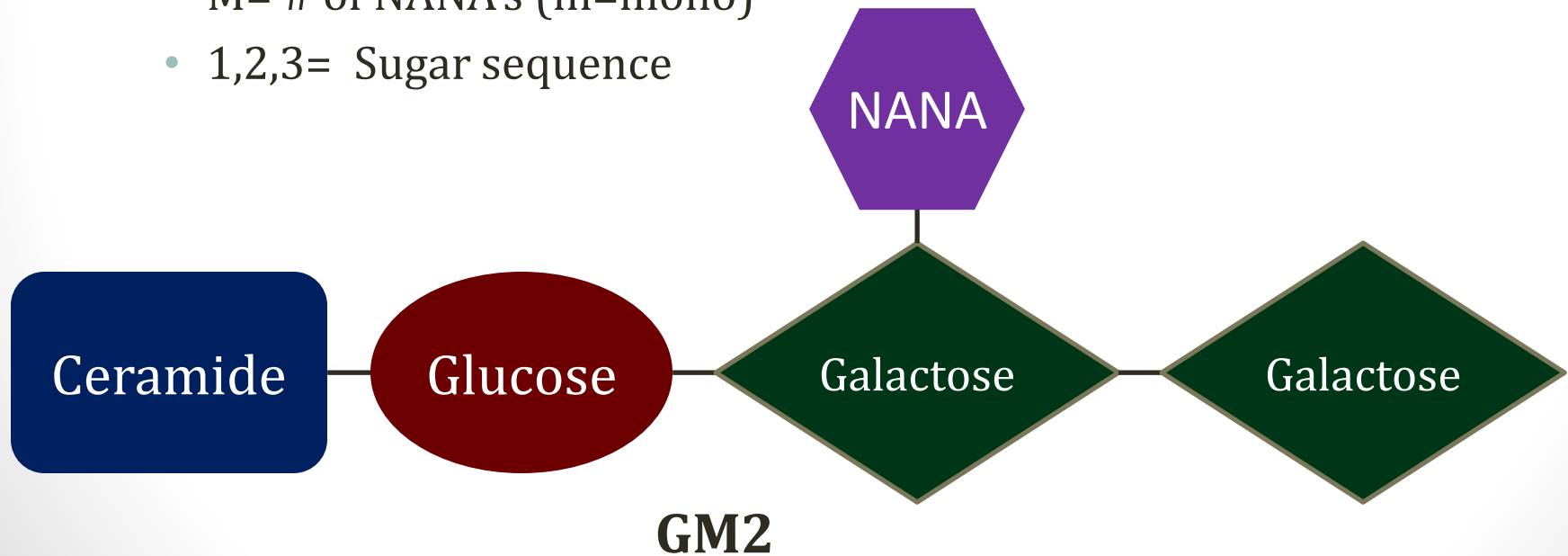
Globoid Cells

- Krabbe: globoid cell leukodystrophy
- Globoid cells in neuronal tissue
 - Globe-shaped cells
 - Often more than one nucleus



Gangliosides

- Contain head group with NANA
 - N-acetylneuraminic acid (also called sialic acid)
- Names GM1, GM2, GM3
 - G-ganglioside
 - M= # of NANA's (m=mono)
 - 1,2,3= Sugar sequence



Tay-Sachs Disease

- Deficiency of hexosaminidase A
 - Breaks down down GM2 ganglioside
- Accumulation of GM2 ganglioside
- More common among **Ashkenazi Jewish** population

Tay-Sachs Disease

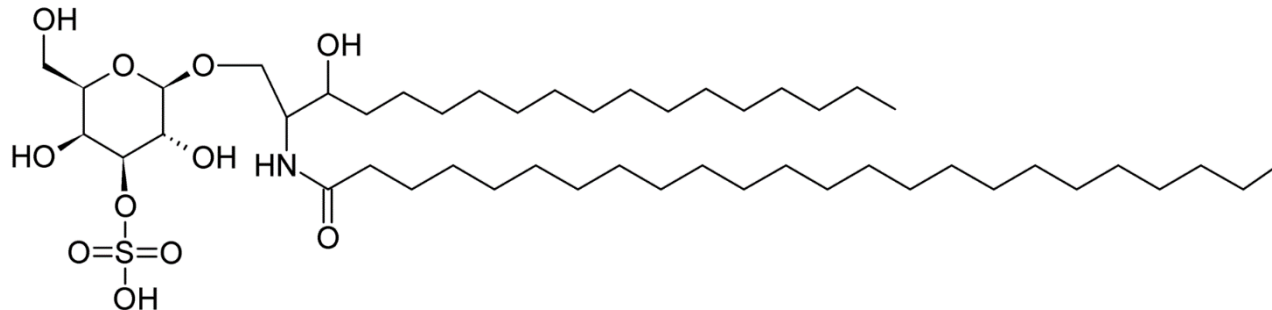
- Most common form presents 3-6 months of age
- Progressive neurodegeneration
 - Slow development
 - **Weakness**
 - Exaggerated startle reaction
 - Progresses to **seizures, vision/hearing loss, paralysis**
 - Usually death in childhood
- Cherry red spot
 - No hepatosplenomegaly (contrast with Niemann-Pick)
- Classic path finding: lysosomes with **onion skinning**

Tay-Sachs Disease

- Classic presentation:
 - 3-6 month old infant
 - Ashkenazi Jewish descent
 - Developmental delay
 - Exaggerated startle response
 - Cherry Red spot

Deficiency of hexosaminidase A
Accumulation of GM2 ganglioside

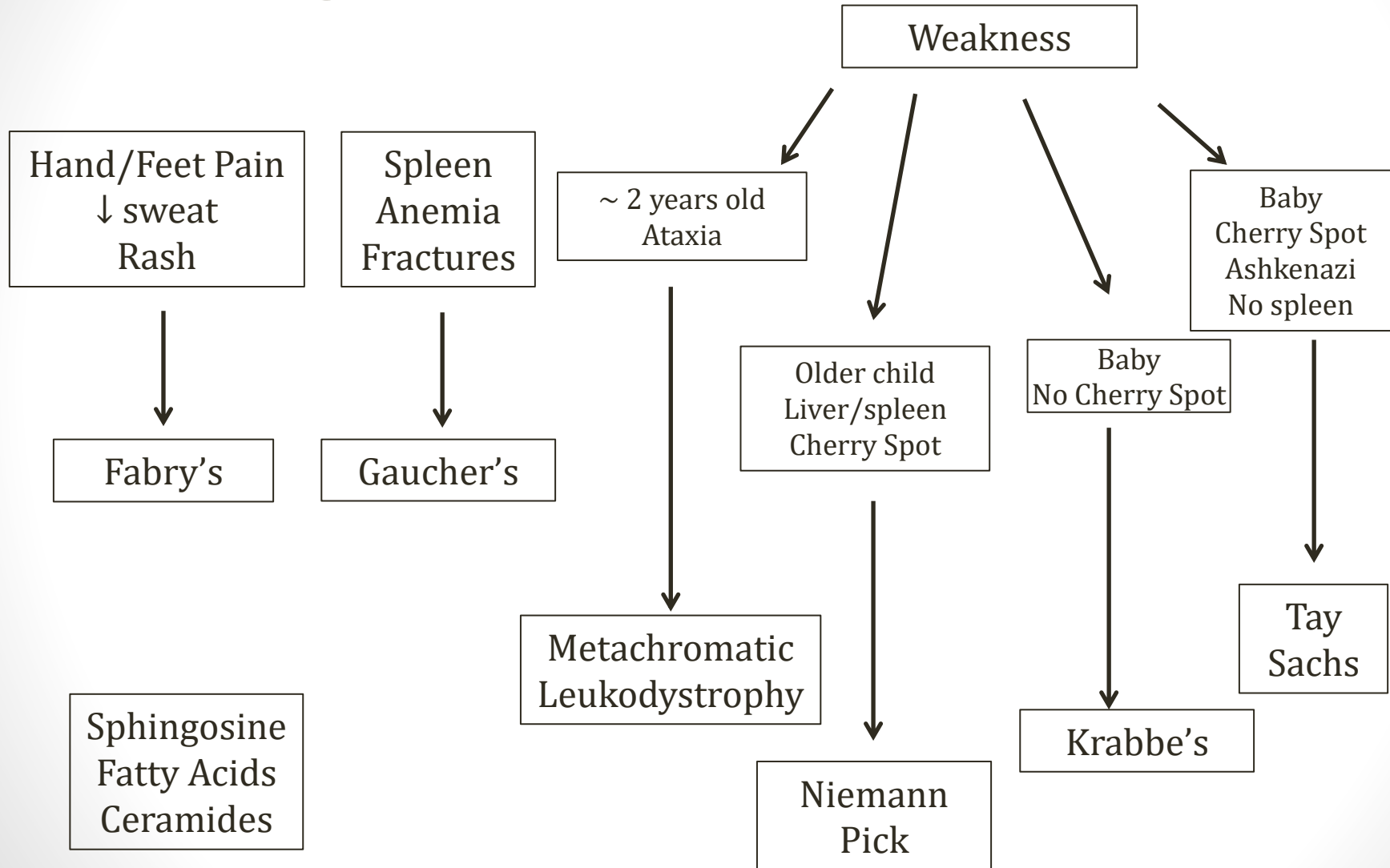
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Metachromatic leukodystrophy

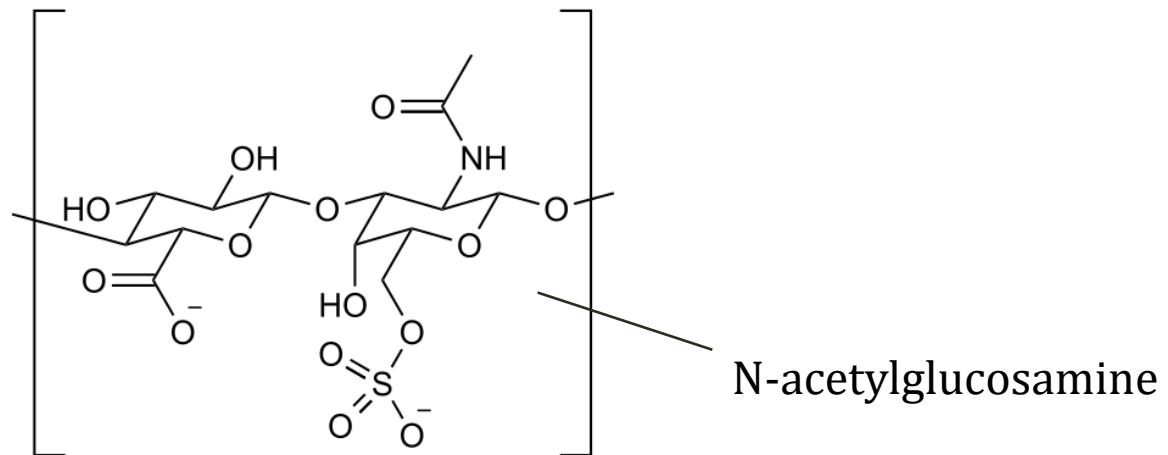
- Autosomal recessive
- Childhood to adult onset based on subtype
 - Most common type presents ~ 2 years of age
 - Contrast with Krabbe's: present < 6 months
- Ataxia: Gait problems; falls
- Hypotonia: Speech problems
- Dementia can develop
- Most children do not survive childhood

Sphingolipidoses



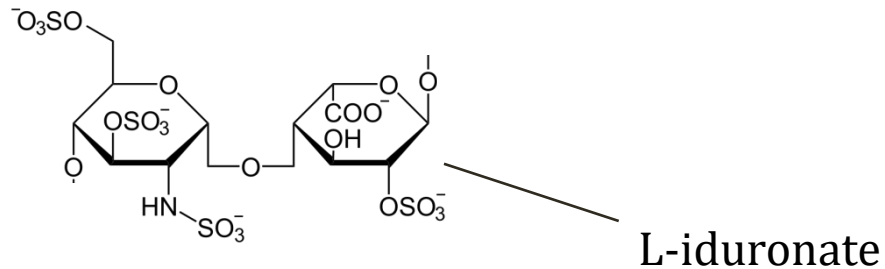
Glycosaminoglycans

- Also called mucopolysaccharides
- Long polysaccharides
- Repeating disaccharide units
- An amino sugar and an uronic acid

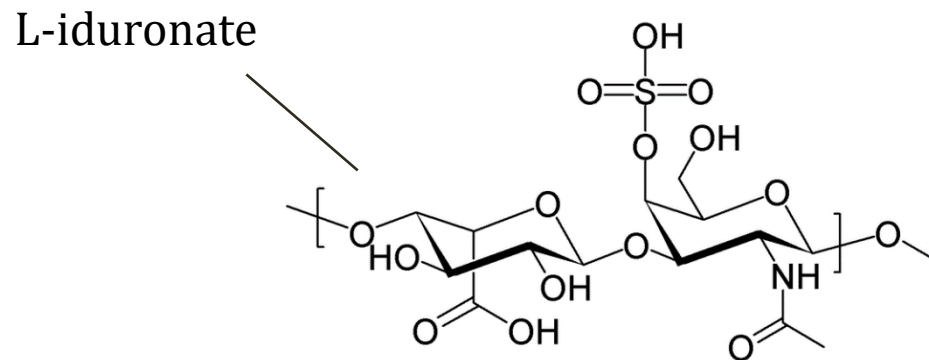


Chondroitin Sulfate

Glycosaminoglycans



Heparan Sulfate



Dermatan Sulfate

Hurler's and Hunter's

- Metabolic disorders
- Inability to breakdown heparan and dermatan
- Diagnosis: mucopolysaccharides in urine
- Types of mucopolysaccharidosis
 - Hurler's: Type I
 - Hunter's: Type II
 - Total of 7 types

Hurler's Syndrome

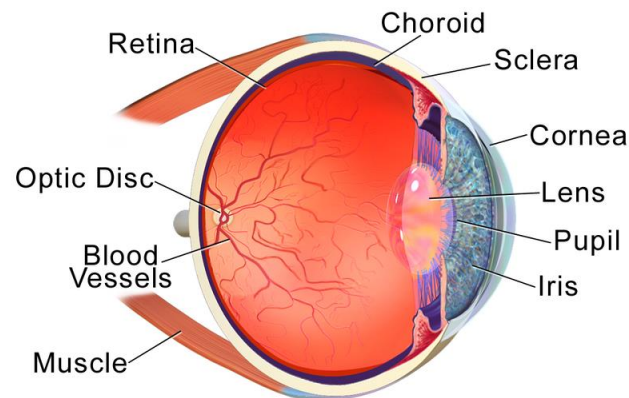
- Autosomal recessive
- Deficiency of **α -L-iduronidase**
- Accumulation of **heparan and dermatan sulfate**
- Symptoms usually in 1st year of life
- Facial abnormalities (“coarse” features)
- Short stature
- Mental retardation
- Hepatosplenomegaly

Dysostosis Multiplex

- Radiographic findings in Hurler's
- Enlarged skull
- Abnormal ribs, spine

Hurler's Syndrome

- **Corneal clouding**
 - Abnormal size arrangement of collagen fibers
- Ear, sinus, pulmonary **infections**
 - Thick secretions
- Airway obstruction and sleep apnea
 - Tracheal cartilage abnormalities



Anatomy of the Eye

BruceBlaus/Wikipedia

Hunter's Syndrome

- X-linked recessive
- Deficiency of **iduronate 2-sulfatase** (IDS)
- Similar to Hurler's except:
 - Later onset (1-2years)
 - No corneal clouding
 - Behavioral problems
 - Learning difficulty
 - Trouble sitting still (can mimic ADHD)
 - Often aggressive behavior

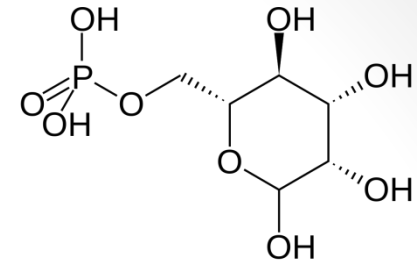
I-cell Disease

Inclusion Cell Disease

- Subtype of **mucopolysaccharidosis** disorders
 - Combined features of sphingolipid and mucopolysaccharide
- Named for inclusions on light microscopy
- Similar to Hurler's
 - Onset in 1st year of life (some features present at birth)
 - Growth failure
 - Coarse facial features
 - Hypotonia/Motor delay
 - Frequent respiratory infections
 - Clouded corneas
 - Joint abnormalities
 - Dysostosis multiplex

I-cell Disease

Inclusion Cell Disease



Mannose-6-Phosphate

- Lysosomal enzymes synthesized normally
- Failure of processing in **Golgi** apparatus
 - **Mannose-6-phosphate** NOT added to lysosome proteins
 - M6P directs enzymes to lysosome
 - Result: enzymes secreted outside of cell
- Key findings:
 - Deficient intracellular enzyme levels (WBCs, fibroblasts)
 - Increased extracellular enzyme levels (plasma)
 - **Multiple** enzymes abnormal
 - Intracellular **inclusions** in lymphocytes and fibroblasts

Pompe's Disease

Glycogen Storage Disease Type II

- Acid alpha-glucosidase deficiency
 - Also “lysosomal acid maltase”
- Accumulation of glycogen in lysosomes
- Classic form presents in infancy
- Severe disease → often death in infancy/childhood